Class-Specific Autoaugment Architecture Based on Schmidt Mathematical Theory for Imbalanced Hyperspectral Classification

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***Abstract*— Hyperspectral image classification (HSIC) often suffers from severe imbalanced category distribution in real applications, which causes bias toward the dominated categories. As an effective method, the deep generative model (DGM) can be used to augment the features of imbalanced data through a learnable method to achieve superior classification performance. However, the features extracted by DGM are preset as a standard Gaussian distribution which results in low interclass difference. Besides, the generated features are too consistent with the original ones, which cannot play a positive role in the discriminability of minority categories (MCs). To conquer these drawbacks, we propose a class-specific autoaugment architecture based on Schmidt mathematical theory (CACS) for the challenging of imbalanced data which consists of two stages: one is training a superior features extractor, and the other one is augmenting features. The class-specific features of the whole HSI are extracted in stage one that supports the following feature augmented module. Specifically, we weighted the classifier in the first phase according to cost-sensitive learning, to prevent the classifier from overfitting. To expand the dispersion between categories, we construct feature prototypes obeying different Gaussian distributions for each class, respectively, and generate class-specific features. Then, the features are augmented in the second phase based on Schmidt's mathematical theory, which enhances the discriminability of minority class features, thus further improving the classification accuracy with interpretability. Extensive experimental results on three benchmarking datasets demonstrate that CACS is outstanding in comparison algorithms, especially in MCs.**

***Index Terms*— Imbalanced data, interclass difference, Schmidt mathematical theory, weight balance.**

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I. Introduction

HYPERSPECTRAL imaging technology can capture spectrum reflected or radiated from materials over hundreds of continuous narrow bands [1]. Among various hyperspectral image (HSI)-processing technologies, HSI classification (HSIC) aims to distinguish land cover categories with slight differences, by recognizing the unique and fine spectral information of different substances in HSIs [2]. Over the last decade, researchers have been constantly innovating various algorithms to upgrade the accuracy of HSIC [3].

With the flourishing of deep learning (DL), HSIC methods based on DL have attracted wide attention. Among various neural networks, the methods based on convolutional neural network (CNN) achieved encouraging results, as CNNs can extract rich deep information which made its performance far superior to the traditional methods such as support vector machine (SVM) [4]. Specifically, 1-D-CNN is applied to HSIC that can only extract spectral dimension information. Then, researchers utilized 2-D-CNN [5] to extract the spatial information of HSIs. To mine rich spatial–spectral information and further improve classification accuracy, 3-D-CNN [6] was introduced into HSIC simultaneously. However, with the complexity of 3-D-CNN increasing, overfitting became more serious. To tackle this problem, Zhong et al. [7] proposed a spatial–spectral residual network (SSRN) that fully extracted the robust features from HSIs, meanwhile preventing the deep network from overfitting by residual blocks connecting. In addition, Roy et al. [8] proposed HybridSN, which cleverly combined 3-D-CNN and 2-D-CNN to reduce the complexity of the network while still extracting HSI information excellently.

While a multitude of HSIC algorithms has achieved certain performance improvements, they ignored that HSIs suffered from an imbalanced proportion of different ground objects, and the minority classes often contain more significant information [9], [10]. For example, in the surveys on crop pests and forest fires, the anomalies only occupy small regions

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in the initial phase but deserve more attention. Imbalanced data refers to the unbalanced distribution of sample quantity. When imbalanced data is input into the classifier, the samples belonging to minority categories (MCs) will have less weight in calculating classification loss, which will result in the models trained on imbalanced data tending to overfit the majority (dominant) classes. That is to say, although the overall accuracy (OA) is satisfactory, the classification effect in the MC is poor [11], [12]. In addition, HSIs are affected by the incident angle, atmospheric interference, unpredictable occlusion, and so on, resulting in spectral distortion that is reflected in two main difficulties [2], [13]. On the one hand, high intraclass variability makes it difficult for the classifier to identify the specified class. On the other hand, low interclass diversity makes it difficult to distinguish between different classes. Therefore, all these defects lead to HSIC still being a challenging task and meriting extensive research.

To address this complicated issue, various classification strategies for imbalanced natural images can be applied to imbalanced HSIs. The solutions can generally be divided into two types: One is data augmentation that synthesizes samples of minority classes to enhance the diversity and sufficiency of training data, including basic data augmentation and deep generative model (DGM) [14]. The former is a fixed way to augment data in the input space. While the latter learns a mapping from the latent distribution to the data space, so the augmented data can better fit the actual distribution. As a representative deep generate algorithm, generative adversarial net (GAN) [15] searches Nash equilibrium in minimax games to train an outstanding generator and classifier. Since features are initialized completely randomly, it is difficult to find the optimal solution in a short time. Another typical deep generative model, variational autoencoder (VAE) [16] applied variational inference to reduce the distance between different distributions. However, VAE only multiplies through replicating the features of minority classes, which makes the generated features cannot provide additional information for the classifier. Moreover, it is assumed that the posterior distribution of all features follows the standard Gaussian distribution, which narrows the distinction between features of different classes.

The other approach is the cost-sensitive algorithm that reweights the proportion of different categories in the loss function of the classifier, that is, the significance weight coefficients of dominated categories are lower than those of MCs [17], [18]. The weighted classifier pays more attention to the distribution of minority classes to solve the problem of imbalanced data. At present, this method is rarely used to deal with imbalanced HSIs. Nevertheless, this method only takes into account the global number of categories and sets equal weight for samples from the same category, which ignores the fact that samples within the same category have different classification difficulties, and some non-MCs may be difficult to classify accurately.

To conquer the aforementioned drawbacks, we proposed a class-specific autoaugment architecture based on Schmidt mathematical theory (CACS) that is divided into two stages. In the first stage, CACS extracts the specific category features of the whole HSI to obtain as much information on MCs as possible to construct more accurate orthogonal projection spaces when augmenting features based on the Schmidt mathematical theory in the second stage. Specifically, to extract the high-quality features of the whole HSI, we use supervised learning to train the 3-D encoder, 3-D decoder, and a custom imbalanced classifier in an end-to-end fashion during the first stage. Since the classifier may pay much attention to the dominant class when classifying imbalanced features, we weighted the classifier based on the cost-sensitive loss function, in which the features are weighted according to their impact on the classifier's decision boundary. In addition, we construct feature prototypes that obey different Gaussian distributions for each class, respectively, and utilize the maximum mean difference (MMD) loss to constrain the features fitting to the corresponding class prototypes. By this means, we enlarge the interclass difference and enhance the intraclass compactness to address the spectral variation issue of HSIs. In the second phase, CACS maps class-specific features to the corresponding Schmidt spaces and learns to select the features with high confidence as the augmented features to obtain the support of mathematical theory and promote interpretability. Therefore, the augmented features maintain the category distinction, increase the discriminability, and avoid the generated samples falling into the local optimal solution, and the classification accuracy of MCs is improved. In particular, in the process of generating features of MCs based on Schmidt's mathematical theory, features of each class are orthogonally projected separately. If feature extraction and generation are completed in one stage, and the number of MCs is too small or even nonexistent in each batch, resulting in the obtained dimension of Schmidt space is too low to express the characteristics of the class. Therefore, we extract all features in HSIs in the first stage and then calculate the corresponding Schmidt space of each category through all feature vectors, which will avoid contingency and be more accurate. In brief, the main contributions of this article can be summarized as follows.

1) We propose a class-specific autoaugment classifier (based on mathematical theory) for HSIs, which will enable us to differentiate between the original features of different categories. Specifically, the model forces samples of the same class to its class prototype in the latent space and pulls samples of different categories apart from the rest of the class prototype.
2) To enrich the diversity of generated samples, instead of resampling the minority class features, we generate the minority samples based on Schmidt's mathematical theory, which theoretically promotes the interpretability of CACS.
3) In addition, we modify the loss weight of the imbalanced classifier in the first stage to reduce the over-fitting of decision boundaries and make the classifier more robust. Based on ten independent tests across three benchmark datasets, the proposed CACS can improve performance over current state-of-the-art methods, including higher average precision and lower deviation.

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The article is organized as follows. In Section II, we briefly introduce the related works. Section III describes the detail of our proposed method. The experimental results and analysis are presented in Section IV. Section V shows the conclusion.

II. Related Works

A. Basic Image Augmentation

In recent decades, a strategy based on original image processing was first proposed. The common methods include rotation, scaling, flipping, and cropping [19]. But drawbacks existed, for instance, some parts of the image may be moved out or not realistic by rotation or flipping and resolution may be affected by zooming [20]. Image erasure technology, such as Cutout, replaces one or more regions in an image with constants. For example, Random erasing proposed by Zhong et al. [21] simply selected a rectangular region in an image at random and replaced it with random numbers, which can greatly improve the result. This is because there is much redundant information in the image. By covering a certain proportion of the image, the risk of overfitting can be reduced and the generalization ability of the model can be improved [22].

Additionally, as another basic image enhancement method, the synthetic minority oversampling technique (SMOTE) synthetized new samples for categories with few samples based on K-nearest neighbor, but the generated samples are not general enough, which can easily result in overfitting. Li et al. [23] proposed a resampling method based on orthogonal complement subspace projection (OCSP), which oversampled minority classes and undersampled majority classes. However, this method was aimed at relatively low-dimensional spectral data, not a 3-D HSI patch.

B. Deep Generative Model

Recently, instead of augmenting data manually through fixed methods, researchers sought an automatic method to generate data efficiently [24]. The DGM extracts the feature distribution from the potential space of the dataset and automatically generates features of minority classes by resampling strategy. DGM can learn which of the generated samples really help the classifier train [25]. The existing methods can be roughly divided into two types: one is based on GANs, and the other is based on the autoencoder [26].

1) DGM Based on GANs: GANs [27] alternately trained a generator and a discriminator so that the generator can create samples that are real enough to confuse the discriminator, and the discriminator can also determine whether the samples are realistic or not. Then, the GAN was applied to the research of HSIC. To make full use of the spatial and spectral information in HSIs, Zhu et al. [15] applied 3-D-CNN to GANs and proposed 3-D auxiliary classifier GAN (3D-ACGAN), in which the discriminator could not only distinguish categories, but also distinguish the truth and falseness of generated samples. Roy et al. [28] proposed a generative adversarial minority oversampling for HSIC (HyperGAMO) in which they alternately trained a convex 3-D patch generator, a 3-D patch extractor, a 3-D conditional discriminator, a 3-D classifier network, and a conditional feature mapping unit. Sample generation and classifier construction are completed in one stage. However, the initial potential vector inputting to the generator is random noise, which complicates the training of GANs.

2) DGM Based on the Autoencoder: As an unsupervised learning algorithm, the autoencoder includes an encoder $\mathbf{z} = q(\mathbf{x})$ for extracting features and a decoder $\mathbf{r} = p(\mathbf{z})$ for generating reconstructions to approximate the output to the input. A network that can only achieve $p(q(\mathbf{x})) = \mathbf{x}$ does not make sense. To tackle this problem, VAE introduced Gaussian noise $\boldsymbol{\sigma}^2$ between the encoder and the decoder, so that the reconstructed vectors of the decoder are robust to noise. Specifically, via the Bayes rule, VAE assumes that the probability to produce picture X is $p(\mathbf{x})$. Real sample $\mathbf{x}$ is mapped to Gaussian distribution $\mathbf{z}$ in feature space by the 3-D encoder, which is expressed as $q(\mathbf{z}|\mathbf{x})$, and the corresponding 3-D decoder is expressed as $p(\mathbf{x}|\mathbf{z})$. Through the reparameterization trick, features $p(\mathbf{z})$ are sampled and then calculated the predicted samples by the 3-D decoder. Minimize the error between the reconstructed samples and the real samples, when the samples X are input

$$\max L = \sum \log p(\mathbf{x}) \tag{1}$$

where

$$p(\mathbf{x}) = \int_{\mathbf{z}}^{\mathbf{x}} p(\mathbf{z}) q(\mathbf{x}|\mathbf{z}) d\mathbf{z}. \tag{2}$$

After a series of deductions, the evidence lower bound (ELBO) of the loss function can be calculated as

$$\mathcal{L}(\theta, \phi; \mathbf{x}) = -D_{KL}\left(q_\phi(\mathbf{z}|\mathbf{x}) \| p_\theta(\mathbf{z})\right) + \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{z}|\mathbf{x})\right] \tag{3}$$

where θ and ϕ are the training parameters of the encoder and the decoder, respectively, and D_{KL} represents the Kullback–Leibler (KL) divergence.

To further improve the accuracy, Sohn et al. [29] added annotated information with training data to assist in sample generation and proposed conditional VAE (CVAE). Different from VAE, CVAE concatenated training data with labels and then input them into the encoder. Notably, no separate loss function is calculated for labels, so CVAE still belongs to the autoencoder. The ELBO of its loss function can be calculated as

$$\mathcal{L}(\theta, \phi; \mathbf{x}, \mathbf{y}) = -D_{KL}\left(q_\phi(\mathbf{z}|\mathbf{x}, \mathbf{y}) \| p_\theta(\mathbf{z}|\mathbf{x})\right) + \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x},\mathbf{y})}\left[\log p_\theta(\mathbf{z}|\mathbf{x}, \mathbf{y})\right]. \tag{4}$$

Wang et al. [30] proposed a newly generated model for HSIC called CVAE with adversarial training (CVA2E), which takes into account the inherent characteristics of spectral when applying deep generative model to HSIs, and combined conditional variational reasoning with adversarial training in spectral sample generation. However, when the generated vectors belong to minority classes, the generated samples may overlap with the real samples. In this case, it is difficult to ensure that the generated ones will help train the classifier. Meanwhile, samples of majority categories are undersampled,

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which wastes the information of these samples and sacrifices the accuracy of dominated categories.

On the basis of VAE, from the perspective of optimal transport (OT), Kolouri et al. [31] proposed the Wasserstein autoencoder (WAE) that changed the calculation method of the reconstruction error. VAE matched $\int q(\mathbf{z}|\mathbf{x})dp(\mathbf{x})$ with $p(\mathbf{x})$, while WAE forced actual distribution $p(\mathbf{z})$ to match predict distribution $q(\mathbf{z}|\mathbf{x})$ by Wasserstein distance, which was a more accurate measure of the similarity between distributions. To some extent, it can solve the pattern collapse problem. Similar to VAE, the loss function of WAE consists of two components: reconstruction loss and prior loss of $\mathbf{z}$

$$\mathcal{L}(\theta, \phi; \mathbf{x}) = \inf_{q_\phi(\mathbf{z}|\mathbf{x}) \in \mathcal{Q}} \mathbb{E}_{p_\mathbf{x}} \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[c(\mathbf{x}, \hat{\mathbf{x}})] + \lambda \cdot \mathcal{D}_\mathbf{z}\big(q_\phi(\mathbf{z}|\mathbf{x}), p(\mathbf{z})\big) \quad (5)$$

where c measures the distance between $\mathbf{x}$ and $\hat{\mathbf{x}}$, and inf represent the minimum certainty.

Combined with CVAE and WAE, Wang et al. [14] proposed a [deep generative classifier (DGC)], which implemented data perturbation (generate features of minority classes) and classification training together. Just like WAE, DGC made the marginal distribution $q(\mathbf{z})$ of $q(\mathbf{z}|\mathbf{x}, \mathbf{y})$ match the prior distribution $p(\mathbf{z})$. The concrete loss function of DGC is as

$$\mathcal{L}(\theta, \phi; \mathbf{x}, \mathbf{y}) = \inf_{q_\phi(\mathbf{z}|\mathbf{x}, \mathbf{y}) \in \mathcal{Q}} \mathbb{E}_{p_\mathbf{x}} \mathbb{E}_{p_\mathbf{y}} \mathbb{E}_{q_\phi(\mathbf{z_i}|\mathbf{x}, \mathbf{y})} \big[c_1\big(\mathbf{x}, \hat{\mathbf{x}}\big) + c_2\big(\mathbf{y}, \hat{\mathbf{y}}\big)\big] + \lambda \cdot D_\mathbf{z}\big(q_\phi(\mathbf{z}|\mathbf{x}, \mathbf{y}), p(\mathbf{z})\big). \quad (6)$$

Among it, c_1 measures the distance between $\mathbf{x}$ and $\hat{\mathbf{x}}$ via $c_1(\mathbf{x}, \hat{\mathbf{x}}) = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2$, and c_2 calculates the difference between label $\mathbf{y}$ and predict label $\hat{\mathbf{y}}$ via cross-entropy loss function $c_2(\mathbf{y}, \hat{\mathbf{y}}) = \mathbf{y} \log \hat{\mathbf{y}} + (1 - \mathbf{y}) \log(1 - \hat{\mathbf{y}})$. DGC presets data of all categories following a single Gaussian distribution in the latent space. It introduces the annotated information into the training data so that the network can learn the corresponding feature distribution of each category by itself. However, the label information cannot be fully utilized only by concatenating labels and data. Therefore, the feature distribution may not be different according to the category. DGC cannot generate prediction data with sharp edges when HSI samples belong to a mixed distribution.

Our proposed CACS constructs different feature prototypes for each category that obey normal distribution. By constraining MMD loss, the distribution of the trained feature of the jth category $\mathbf{z}_j$ fits the distribution of the jth prototype, which ensures class specificity in the latent space. In addition, DGC simply multiplies the features of MCs, which can only superficially resolve the category imbalance from the number of samples, and the generated features contain no additional information. On the basis of Schmidt's mathematical theory, CACS augments the features of MCs to enhance the discriminability of features.

III. Proposed Method

In this section, we mainly introduce the basic structure of CACS for imbalanced HSIC. First, we clarify the architecture of the two-stage model and the detail of weighting the classifier in the first stage. Then, the process of generating class-specific latent vectors is described. Finally, we interpret how CACS augments features with rich interclass diversity.

A. Frameworks of the CACS

As shown in Fig. 1, the first stage of our proposed CACS mainly consists of the 3-D encoder, the 3-D decoder, and the custom imbalanced classifier. In this stage, class-specific features are extracted from the entire HSI. In the second stage, the feature generator based on the Schmidt mathematical theory projects all MCs into the corresponding Schmidt space. Therefore, reconstructing features based on Schmidt's mathematical theory are more accurate because they contain as much information on MCs as possible. If the features are extracted and generated in one stage, the minority samples in each training batch are few or even nonexistent. As a result, the dimensions of Schmidt space projected by the MC are too low, and the basis vectors of the Schmidt space cannot express the features of the MC, which makes the generated vectors biased to one-sided information. The specific flow of CACS is as follows.

The HSI with C classes expressed as $P \in R^{H \times W \times L}$ was first segmented to N spatial neighborhood image patches $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$, where $\mathbf{x}_i \in R^{m \times m \times L}$, $i \in (1, N)$. The label of the center pixel $\mathbf{Y} = \{\mathbf{y}_1, \ldots, \mathbf{y}_C\}$ is denoted as the label of the whole patch. The training units are 3-D neighborhood image patches rather than single spectra so that spectral–spatial features in HSIs can be simultaneously extracted. In the first stage, imbalanced 3-D HSI patches $\mathbf{X}$ are mapped to latent spaces by the 3-D encoder Enc. Then, the extracted features are forced to be constrained to the corresponding feature prototypes $\mathbf{L} = \{\mathbf{l}_1, \ldots, \mathbf{l}_C\}$, to obtain the class-specific features $\mathbf{Z} = \{\mathbf{z}_1, \ldots, \mathbf{z}_N\}$. Next, the features $\mathbf{Z}$ are fed into the 3-D decoder to reconstruct virtual hyperspectral patches $\hat{\mathbf{X}}$ and input to the custom weighted classifier to predict the categories $\hat{\mathbf{Y}}$. Notably, we leverage reconstruction loss and classification loss to constrain the category-specific features $\mathbf{Z}$.

In the second stage, $\mathbf{Z}$ are grouped according to categories, among which the categories whose samples are less than the average N/C are MCs. The category with samples greater than N/C is the majority category, and they are undersampled, as referred to [32]. We extend $\mathbf{Z}$ of MCs based on Schmidt's mathematical theory and generate $\mathbf{Z}_{\text{aug}} = \{\mathbf{z}_{1,1}, \ldots, \mathbf{z}_{1,N/C}, \ldots, \mathbf{z}_{C,1}, \ldots, \mathbf{z}_{C,N/C}\}$. Notably, it is trained in a learnable way to get the appropriate features. We input the augmented features into the classifier and constrain the cross-entropy loss between the predicted labels and the original labels, to improve the classification effect. By constraining the MMD loss between the distribution of generated features $\mathbf{G}_j$ and the distribution of features prototypes $\mathbf{l}_j$, the generated features have class specificity and improve their characterization ability. Compared with DGC, we do not simply replicate the features, but select the features with strong representation ability and excellent classification effect from each Schmidt space. So, the generated features not only retain

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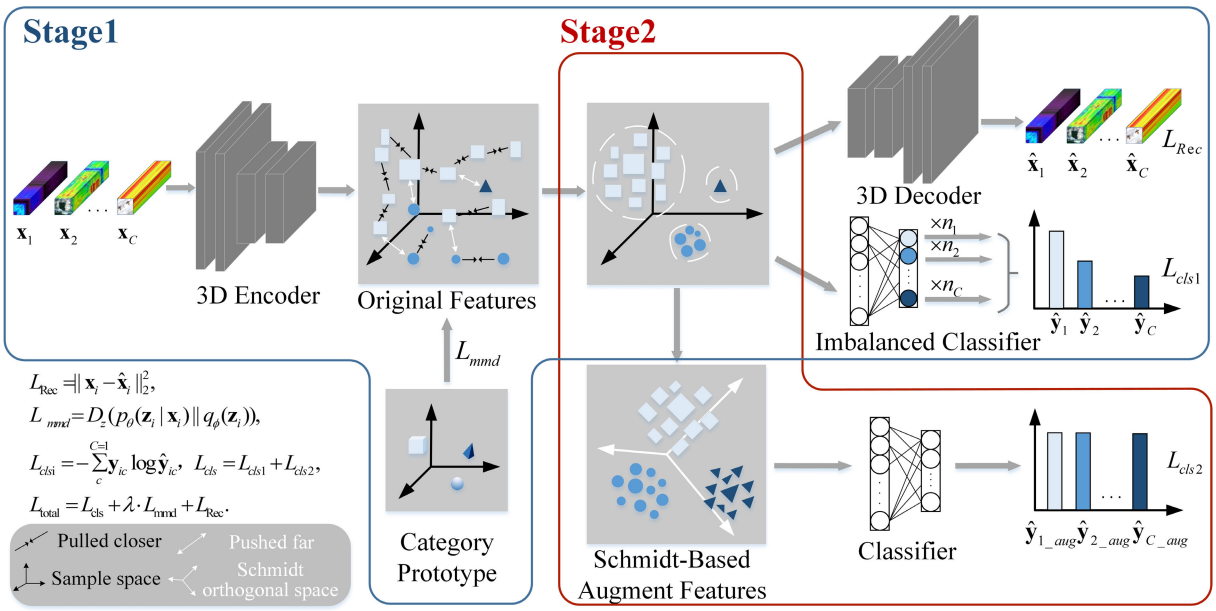


Fig. 1. CACS structure diagram is divided into two stages. The first stage of the model, which is in the box of the blue line, trains the 3-D encoder, the 3-D decoder, and the custom imbalanced classifier in an end-to-end fashion to generate class-specific features. In the second stage, inside the red line, the extracted features were augmented based on Schmidt's mathematical theory.

class-specific semantic information, but also contain novelty, which improves the robustness of the network. Finally, feature $\mathbf{Z}_{\text{aug}}$ is input into the balanced classifier $\mathbf{cls}_{\text{balanced}}$, which uses a fully connected (FC) layer to parse the semantic information in $\mathbf{Z}_{\text{aug}}$ comprehensively, and output predicted label $\hat{\mathbf{Y}}_{\text{aug}} = \{\hat{\mathbf{y}}_{1_{\text{aug}}}, \ldots, \hat{\mathbf{y}}_{C_{\text{aug}}}\}$.

1) 3-D Encoder and 3-D Decoder: The 3-D encoder of CACS consists of a two-stage 3-D CNN module with residual connection to fully extract spectral–spatial features of HSIs, whose structure is shown in Fig. 2. Due to the redundancy of spectral dimension information, the spectral features of his cubes are first extracted by continuous convolution layers with a convolution kernel size of (1,1,7) to reduce the spectral dimension depth. Then, extracts the spatial features of HSI cubes through the continuous convolution with a convolution kernel size of (3,3,1), according to [7]. We adopt the residual connections in both stages to prevent overfitting and the pooling layers to expand the perception area and enhance the invariance of features. It is followed by an FC layer that fixes the feature to zdim at the end. For the ith 3-D-patch $\mathbf{x}_{j,i}$ in class j, we extract the feature with 3-D encoder Enc, the probability of obtaining $\mathbf{z}_{j,i}$ is as

$$P\left(\text{Enc}\left(\mathbf{x}_{j,i}\right) = \mathbf{z}_{j,i}\right) = q_\phi\left(\mathbf{z}_{j,i}|\mathbf{x}_{j,i}\right). \tag{7}$$

The 3-D decoder is mirror symmetric with the 3-D encoder, and the convolution layer is replaced by the deconvolution layer to generate high-quality reconstructed images [33]. Meanwhile, the probability that $\mathbf{z}_{j,i}$ is reconstructed into $\hat{\mathbf{x}}_{j,i}$ is as

$$P\left(\text{Dec}\left(\mathbf{z}_{j,i}\right) = \hat{\mathbf{x}}_{j,i}\right) = p_\theta\left(\hat{\mathbf{x}}_{j,i}|\mathbf{z}_{j,i}\right). \tag{8}$$

2) Custom Imbalanced Classifier: To enhance the representation of category-specific features, we take full advantage of the labeled data and input the imbalanced features extracted by CACS in the first stage into the classifier. Due to the large quantity gap in each category, the feature distribution is imbalanced. The softmax function in the classifier applies gradient descent to minimize the standard cross-entropy loss, which causes the classifier to bias toward the dominant categories. To refine the network, we adaptively assign different weights to features in the classifier according to the number of samples belonging to the same category and the impact on the decision boundary [34]. The classifier can avoid overfitting to dominated categories or individual samples far away from others on the decision boundary. Specifically, before e epochs, we train the classifier according to the normal cross-entropy loss function. After that, since we are interested in the degree of overfitting of the decision boundary, we focus on the change in the last FC layer of the classifier. We set $f(\mathbf{z}_j, w)$ as the classifier trained by parameter w. According to the influence function [35], it can be seen that the influence of removing point $\mathbf{z}_j$ on model parameters can be estimated by changing parameters in $f(\mathbf{z}_j, w)$ to judge the relationship between the change of classifier parameters and the change of sample weight $\mathbf{z}_j$. We fine-tuned the boundary to weight the features according to the reciprocal of their influence on the classifier's decision boundary as a proportion, which reduces the overfitting of most class features. The importance factor can be written as

$$\text{IF}(\mathbf{z}_{j,i}; w) = \nabla_w L(y_i, f(\mathbf{z}_{j,i}, w))_1. \tag{9}$$

When the classifier applies softmax cross-entropy loss, the loss function is $L(y, f(z_{j,i}, \hat{w})) = -\sum_n^{i=1} y_i \log f_i$, where f_i is the ith output of the classifier corresponding label y_i.

Since we are interested in overfitting the decision boundaries of the model, we focus on the last FC layer of the network. Let the last FC be $w = \{w_1, w_2, \ldots, w_k\}$ and the input vector is expressed as $w = \{t_1, t_2, \ldots, t_k\}$. Then, the importance factor

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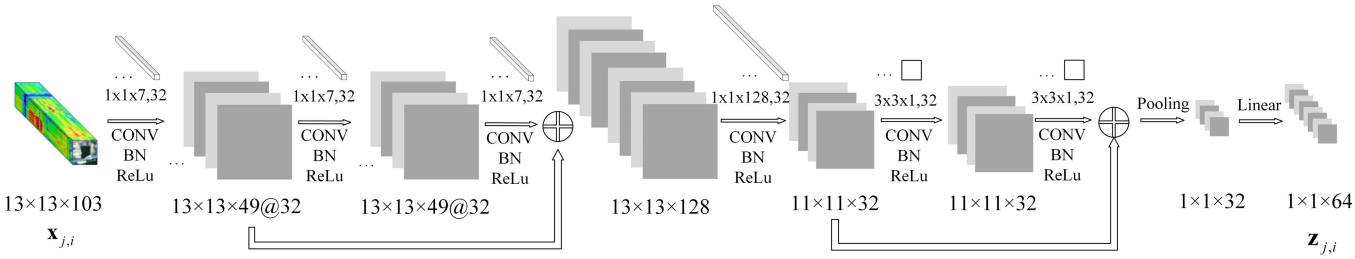


Fig. 2. 3-D encoder extracts spectral and spatial information through a two-stage 3-D CNN with residual connections.

is expressed as

$$\mathrm{IF}(\mathbf{z}_j; w) = \|f(\mathbf{z}_{j,i}, w) - y_i\|_1 \cdot \|t\|_1. \tag{10}$$

The loss function under the influence factor weighting is defined as

$$L_{\mathrm{Reweight}}(y_i, f(z_{i,j}, w)) = \frac{L(y, f(z, w)}{\|f(z, w) - y\|_1 \cdot \|t\|_1}. \tag{11}$$

In addition, we add a class reweighting term λ_k, expanding the weights of MCs. Therefore, the loss function is reweighted by category on the basis of which we take into account both the influence of each point on the decision boundary and the sample number

$$\lambda_k = \alpha n_k^{-1} \Big/ \sum_{k'=1}^{K} n_{k'}^{-1} \tag{12}$$

where n_k is the number of class k of the training data and alpha is the introduced hyperparameter. The influence of this parameter on the network is shown in the ablation study. The final loss function is

$$L_{\mathrm{Reweight}}(w) = \frac{1}{m} \sum_{(x,y)=D_m} \lambda_k \frac{L(y, f(z, w))}{\|f(z, w) - y\|_1 \cdot \|t\|_1}. \tag{13}$$

B. Class-Specific Feature Aggregation

Previously, deep generative models such as VAE [36] and DGC [14] generated new samples from the latent variables, but only assumed that all features obey a single Gaussian distribution in the latent space. Nevertheless, for HSIC, it is crucial to extract the differences between categories to enhance the classification effect, as the differences within categories are much smaller than those between categories. Therefore, the generator divides the features extracted by the encoder into different Gaussian distributions according to the categories and fits the corresponding category of feature prototypes. In this way, we pulled together features belonging to the same categories in the normalized embedding space and separated features from different categories. The 3-D encoder extracts the training image blocks $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$ and gets the original latent vectors $\mathbf{Z}_{\mathrm{org}} = \{\mathbf{z}_{\mathrm{org}_1}, \ldots, \mathbf{z}_{\mathrm{org}_N}\}$. The original features are grouped by category according to labels $\mathbf{Y} = \{\mathbf{y}_1, \ldots, \mathbf{y}_C\}$. Using the original features, CACS learns the mean $\boldsymbol{\mu} = \{\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_C\}$ and variance $\boldsymbol{\sigma} = \{\boldsymbol{\sigma}_1, \ldots, \boldsymbol{\sigma}_C\}$ of the Gaussian distribution followed by prototypes of C classes. By applying the reparameterization trick, the embedding prototype $\mathbf{l}_j$ is calculated by sampling from $\varepsilon_{j,i}$ which follows the standard Gaussian distribution $\mathcal{N}(0, 1)$ and transforming as

$$l_{j,i} = \mu_j(\mathbf{x}_{j,i}, \mathbf{y}_j) + \varepsilon_{j,i} \odot \sigma_j^2(\mathbf{x}_{j,i}\mathbf{y}_j). \tag{14}$$

For each class, $\varepsilon_{j,i}$ is generated with different values and resampled with different $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ to get N class prototypes $\mathbf{L} = \{\mathbf{l}_1, \ldots, \mathbf{l}_C\}$. Then, the sample prototype of each class is extended to calculate the loss. Specifically, multiply $\mathbf{l}_j$ by the number of samples of class j.

The MMD is used to constrain the distance between the original features and the category prototypes so that features of the same categories fit their prototype. By this means, the class-specific latent characteristics are generated, and the specific distance formula is as

$$\mathrm{MMD}[F, Z_j, L_j] := \left\| \frac{1}{n} \sum_{i=1}^{n} f(z_{j,i}) - \frac{1}{m} \sum_{k=1}^{m} f(l_{j,k}) \right\|_{\mathcal{H}}. \tag{15}$$

Compared with other information entropy functions such as cross-entropy and Jenson's Shannon (JS) divergence, MMD can better match high-dimensional distribution information such as HSIs [37]. In addition, MMD can also be combined with subsequent stochastic gradient descent (SGD). The objective function can be expressed as

$$L_{\mathrm{mmd}} = D_{\mathbf{z}}(p_\theta(\mathbf{z}_j \mid \mathbf{x}_j, \mathbf{y}_j) || q_\varphi(\mathbf{l}_j)). \tag{16}$$

Next, category-specific features are generated based on the category prototypes. Since blindly generating the minority features will cause information redundancy and the original features and generated features will be mapped to the Schmidt orthogonal space together, we upsample the original features of MCs and downsample the features of majority categories until their number is equal to the dimension of the potential Schmidt space zdim, and the generation proportion parameter is ar. Ablation studies are described in Section IV-E.

C. Features Generation Based on Schmidt's Mathematical Theory

The existing oversampling models simply replicated the extracted latent variables, such as 3-D convolutional autoencoder (3D-CAE) [38], resulting in a lack of diversity. Based on generating class-specific original features that increase the differences between classes, CACS leverages Schmidt's mathematical theory to generate MC features with intraclass diversity, which enriches the theoretical support of data generation, improves the robustness, and alleviates the collapse of the model to a certain extent.

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Specifically, we calculate the basis vectors of the Schmidt space for each category by category-specific features according to the formula of Schmidt orthogonal transformation. For example, the features of the jth category, $\mathbf{z_j} = \{z_{j,1}, z_{j,2}, \ldots, z_{j,n}\}$, are transformed to calculate the Schmidt orthogonality space of class j. The specific steps of Schmidt's orthogonal transformation are performed as follows.

Step 1: Take any nonzero vector $z_{j,1}$ in the features of class j, and set the basis vector $s_{j,1} = z_{j,1}$.

Step 2: Take any nonzero vector $z_{j,2}$ and remove the projection component of $z_{j,2}$ onto the subspace generated by $s_{j,1}$. Specifically, projecting vector $z_{j,2}$ onto vector $s_{j,1}$ to get $((s_{j,1}, z_{j,2})/(s_{j,1}, s_{j,1}))s_{j,1}$ and subtract from $z_{j,2}$ to get $s_{j,2}$

$$s_{j,2} = z_{j,2} - \frac{(s_{j,1}, z_{j,2})}{(s_{j,1}, s_{j,1})} s_{j,1}. \tag{17}$$

Step 3: Take any nonzero vector $z_{j,3}$ and remove the projection component of $z_{j,3}$ onto the subspace generated by $s_{j,3}$. Projecting vector $z_{j,3}$ onto vector $s_{j,1}$ and $z_{j,2}$ to get $((s_{j,1}, z_{j,3})/(s_{j,1}, s_{j,1}))$ and $((s_{j,2}, z_{j,3})/(s_{j,2}, s_{j,2}))$ and subtract $z_{j,3}$ from the sum of the projections to get $s_{j,3}$

$$s_{j,3} = z_{j,3} - \frac{(s_{j,1}, z_{j,3})}{(s_{j,1}, s_{j,1})} s_{j,1} - \frac{(s_{j,2}, z_{j,3})}{(s_{j,2}, s_{j,2})} s_{j,2} \cdots. \tag{18}$$

Step n: Take any nonzero vector $z_{j,n}$ and remove the projection component of $z_{j,n}$ onto the subspace generated by $s_{j,n-1}$. Projecting vector $z_{j,n}$ onto vector $s_{j,1}, s_{j,2}, \ldots, s_{j,n}$ to get $(s_{j,1}, z_{j,n})/(s_{j,1}, s_{j,1})$, $(s_{j,2}, z_{j,n})/(s_{j,2}, s_{j,2}), \ldots, (s_{j,n-1}, z_{j,n})/(s_{j,n-1}, s_{1,n-1})$ and then $z_{j,n}$ minus the sum of $n-1$ projections to get $s_{j,n}$

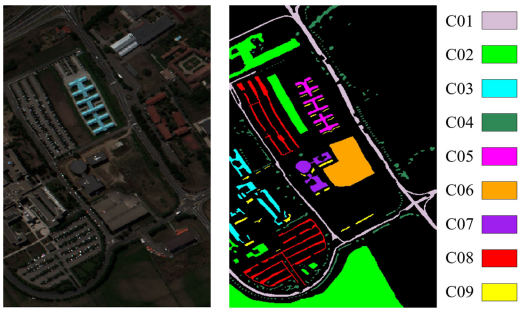
$$s_{j,n} = z_{j,n} - \frac{(s_{j,1}, z_{j,n})}{(s_{j,1}, s_{j,1})} s_{j,1} - \cdots - \frac{(s_{j,n-1}, z_{j,n})}{(s_{j,n-1}, s_{1,n-1})} s_{1,n-1} \tag{19}$$

where $\mathbf{S_j} = \{s_{j,1}, s_{j,2}, \ldots, s_{j,N_j}\}$ is orthogonal basis vectors in Schmidt space of class j, and N_j is the number of latent features of class j. So, the Schmidt orthogonal space of all category features is expressed as $\boldsymbol{S} = \{\boldsymbol{S}_1, \boldsymbol{S}_2, \ldots, \boldsymbol{S}_C\}$, where C is the number of categories.

Then, features are generated in Schmidt spaces of each MC. In particular, the orthogonal basis vectors in Schmidt space of class j are linearly combined to obtain the vector in the same space as the class-specific feature $\mathbf{z_j}$. The specific formula is as

$$\begin{aligned}
\boldsymbol{G}_1 &= \alpha_{1,1} s_{1,1} + \alpha_{1,2} s_{1,2} + \cdots + \alpha_{1,N_1} s_{1,N_1} \\
\boldsymbol{G}_2 &= \alpha_{2,1} s_{2,1} + \alpha_{2,2} s_{2,2} + \cdots + \alpha_{2,N_1} s_{2,N_1} \\
&\cdots \\
\boldsymbol{G}_C &= \alpha_{C,1} s_{C,1} + \alpha_{C,2} s_{C,2} + \cdots + \alpha_{C,N_C} s_{C,N_C}
\end{aligned} \tag{20}$$

where $\alpha_{1,1}, \alpha_{1,2}, \ldots, \alpha_{1,N_1}, \ldots, \alpha_{C,N_C}$ are trainable parameters. The linear correlation between generated features $\mathbf{G_j}$ and $\mathbf{z_j}$ is learned through the network. By constrains MMD loss between the distribution of $\mathbf{G_j}$ and distribution of category prototype $\mathbf{l_j}$, features generated based on Schmidt space maintain category specificity and intraclass consistency, and theoretically promote the interpretability of CACS. It also constrains the cross-entropy loss L_{cls} of the predictive categories of the generated features through the classifier. Through the Schmidt mathematical theory, CACS generates features that both conform to the class distribution and contain intraclass diversity to a certain extent.



(a) (b)

Fig. 3. UP dataset. (a) Pseudo-color composite image (RGB-R:50, G:30, B:20). (b) Ground truth.

TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR NINE LABELS LAND-COVER CATEGORIES IN THE UP DATASET

Class		Number of samples	
No.	Name	Train	Test
1	Asphalt	66	6565
2	Meadow	186	18463
3	Gravel	21	2078
4	Trees	31	3033
5	Painted metal sheets	13	1332
6	Bare Soil	50	4979
7	Bitumen	13	1317
8	Self-Blocking Bricks	37	3645
9	Shadows	9	938
Total		426	42350

IV. EXPERIMENTAL RESULTS AND ANALYSIS

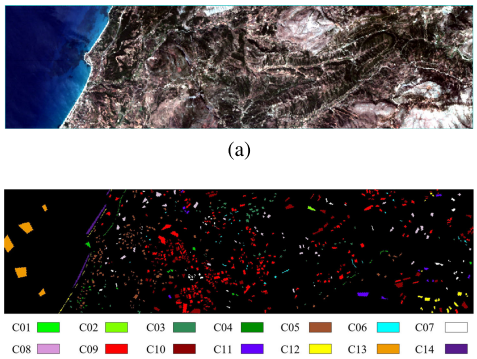
In this section, we verify the performance of the proposed CACS on three benchmark imbalance HSI datasets, including the University of Pavia (UP), Kennedy Space Center (KSC), and Loukia dataset.

A. Experimental Datasets

1) University of Pavia: Captured by the ROSIS sensor from Pavia, UP contains 610 × 340 pixels and the geometric resolution is 1.3 m/ pixel. After removing the low signal-to-noise ratio (SNR) bands, 103 bands remain with the wavelength range of 430–860 nm. Just like the composite pseudo-color image and ground truth shown in Fig. 3(a) and (b), the UP dataset consists of nine land-cover classes. We randomly selected 1% of each class as training samples, whose names and numbers are listed in Table I.

2) Loukia: The next dataset Loukia was shot by the Hyperion sensor. After removing bands absorbed, 176 bands were derived covering the range from 363 to 1018 nm. Besides, the Hyrank training set contains two benchmark HSIs, Dioni and Loukia. In our study, the Loukia dataset was selected as it contains more land-cover categories (14 classes) than Dioni

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(a) (b)

Fig. 4. Loukia dataset. (a) Pseudo-color composite image (RGB-R:22, G:11, B:7). (b) Ground truth.

TABLE II

NUMBER OF TRAINING AND TESTING SAMPLES FOR 14 LABELS LAND-COVER CATEGORIES IN THE LOUKIA DATASET

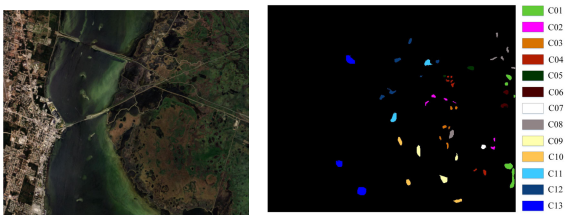
Class		Number of samples	
No.	Name	Train	Test
1	Dense Urban Fabric	14	274
2	Mineral Extraction Sites	3	64
3	Non Irrigated Arable Land	27	515
4	Fruit Trees	4	75
5	Olive Groves	70	1331
6	Broadleaved Forest	11	212
7	Coniferous Forest	25	475
8	Mixed Forest	54	1018
9	Dense Scleorophyllous Vegetation	190	3603
10	Sparce Sclerophyllous Vegetation	140	2663
11	Sparcely Vegetated Areas	20	384
12	Rocks and Sand	24	463
13	Water	70	1323
14	Coastal Water	23	428
Total		675	12828

(12 classes). The pseudo-color composite image and ground truth are shown in Fig. 4(a) and (b). As shown in Table II, 5% samples of each category are randomly chosen as the training set.

3) Kennedy Space Center: The KSC dataset shot by NASA Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) includes 224 bands originally. After water vapor noise bands are removed, there are 176 bands, ranging from 250 to 400 nm. The KSC dataset has 512–614-nm pixels with a geometric resolution is 18 m. Fig. 5(a) and (b) shows the composite false color image and ground truth of KSC. Among the total of 5202 labeled samples, there are 13 classes in the ground truth. In concrete, Table III lists the number of randomly selected 5% training portions from each category.

B. Experimental Setting

We performed ten independent experiments with randomly selected training sets, then averaged the accuracy and variance to get the final result, to test the stability of CACS. The training set is divided into batches with a size of 64, and the dimension of the embedding vector is 64. We measured the network performance based on three criteria metrics: average accuracy (AA), OA, and Kappa coefficient (K). To make CACS fully learn from training data, it was trained for 150 epochs with a learning rate of 0.001. The hardware of the experiment



(a) (b)

Fig. 5. KSC dataset. (a) Pseudo-color composite image (RGB-R:28, G:19, B:10). (b) Ground truth.

TABLE III

NUMBER OF TRAINING AND TESTING SAMPLES FOR 13 LABELS LAND-COVER CATEGORIES IN THE KSC DATASET

Class		Number of samples	
No.	Name	Train	Test
1	Scrub	38	723
2	Willow-swamp	12	231
3	CP-hammock	13	243
4	Slash-pine	13	239
5	Oak-Broadleaf	8	153
6	Hardwood	11	218
7	Swap	5	100
8	Graminoid-marsh	22	409
9	Spartina-marsh	26	494
10	Cattail	20	384
11	Salt-marsh	21	398
12	Mud-flats	25	478
13	Water	46	881
Total		260	4951

is a Geforce GTX 2080Ti server, a 112-GB CPU, and the software is Ubuntu 20.04, python 3.7.0, TensorFlow 1.13.1, Cuda 10.0.13.

C. Comparison With Different Methods

To verify the superiority of the proposed CACS for imbalanced HSIC, we conduct some experiments to compare CACS with other state-of-the-art algorithms, including the traditional DL classifier 3-D-CNN, the representative HSIC network SSRN, and the HybridSN that jointly exploits 3-D-CNN with 2-D-CNN. In addition, CACS is also compared with highly relevant algorithms, for example, 3-D-GAN which also generates latent spatial samples, and HyperGAMO which is the latest generation adversarial minority oversampling network for imbalanced HSIs. We also compare with Adaptive DropBlock-enhanced GANs (ADGANs) [39], which solved the problem that discriminators in Gans always contradict themselves. The training and testing datasets in these comparison experiments are consistent with our CACS.

By the same ten random seeds, 1% from each class in the UP dataset and 5% from each category in the Loukia and KSC datasets were randomly selected as training samples. Since the division of the training and test data in the compared network is different from that of CACS, we slightly modify the input part of the original code. To ensure the reliability of the results, we conducted ten independent repeated experiments

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TABLE IV
CLASSIFICATION ACCURACY OF DIFFERENT ALGORITHMS FOR THE UP DATASET

No.	3D-CNN [6]	SSRN [7]	HybridSN [8]	3D-GAN [15]	HyperGAMO [28]	HC-ADGAN [39]	CACS
01	92.45±1.83	**98.87±2.31**	95.09±2.49	94.49±2.01	95.06±2.43	93.17±2.34	96.96±1.78
02	98.50±0.68	99.75±0.46	99.28±1.32	98.93±0.66	**99.96±0.12**	98.34±0.51	99.41±0.53
03	53.13±6.36	76.31±12.01	79.46±6.43	78.35±9.82	80.39±7.57	**96.52±6.44**	92.82±3.61
04	95.74±1.14	93.72±4.72	90.61±4.68	86.33±3.58	85.55±5.45	91.38±4.23	**96.20±1.50**
05	95.86±3.57	89.54±29.80	99.27±1.54	85.15±2.21	96.06±0.95	99.52±0.68	**99.55±0.36**
06	86.84±3.93	**100.00±0.00**	93.71±2.56	98.30±1.12	92.04±2.47	99.68±2.48	99.87±0.18
07	59.51±8.24	**99.23±1.14**	96.11±3.15	89.79±5.06	95.94±2.20	88.28±2.91	98.26±1.73
08	81.91±4.31	90.00±24.69	80.81±9.11	85.79±2.81	95.96±1.45	97.40±1.93	**97.58±1.99**
09	92.10±6.55	87.44±11.06	89.53±11.27	78.18±7.51	**98.94±1.81**	73.08±2.36	93.77±3.82
AA (%)	84.01±1.79	92.76±3.58	91.54±1.95	88.37±0.81	93.32±1.38	83.79±0.95	**97.16±0.39**
OA (%)	90.90±0.80	96.61±1.88	94.48±1.33	93.95±0.41	96.04±1.27	96.26±0.83	**98.22±0.34**
$k \times 100$	87.87±1.08	95.50±2.48	92.66±1.74	91.97±0.54	94.74±1.67	95.07±1.53	**97.64±0.44**

and calculated the mean and variance of the ten results. The classification accuracies are shown in Tables IV–VI, where the optimal and suboptimal results of each category, AA, OA, and k are bolded and underlined, respectively. We define the class with samples less than the average sample of all classes whose sample size is less than the average sample size as an MC and put bold in the number position to clearly show the classification effect of all algorithms on MCs.

As can be seen from Table IV, for the UP dataset, our proposed CACS achieves the highest AA, OA, and k, and the lowest standard deviations. Because MC features generated by CACS based on Schmidt's mathematical theory contain the category information, the optimal accuracy scores are achieved in MCs 3, 4, 5, and 8 and the suboptimal are achieved in MCs 7 and 9. Especially, the third MC improves by more than 10% compared with the suboptimal method. Additionally, the accuracy of categories 1 and 6 is suboptimal, indicating that CACS improved the accuracy of MCs without sacrificing the classification ability of the dominated classes. It can be analyzed that CACS draws features belonging to the same category closer to the category prototype in latent space so that the network can learn intercategory differences in HSIs.

The experimental results of the Loukia dataset are shown in Table V. Compared with the other two datasets, the accuracies of all methods are relatively low. Perhaps, the Loukia dataset was taken by a spaceborne hyperspectral camera, resulting in a lower resolution than the UP and KSC datasets taken by airborne hyperspectral cameras. Even under this condition, the OA, AA, and k of CACS are still superior to other comparison algorithms, and the classification accuracies are the best in MCs 2 and 3, and the suboptimal in MCs 4, 6, and 7. Notably, in MCs 2 and 3, it outperformed other comparison algorithms by tens of percentage points. Furthermore, HyperGAMO oversampled samples of MCs in the latent space through a convex 3-D patch generator, so it acquires the optimal accuracy in MCs 1, 4, 6, 7, 11, and 12. However, the samples generated by HyperGAMO may be limited to a narrow range and cannot accurately represent the distributions of all MCs, which results in the classification results of MCs 3, 4, and 6 being too low. Besides, it severely sacrifices the accuracies of majority classes when oversampling MCs, which results in the lowest scores for categories 5, 8, 10, 13, and 14.

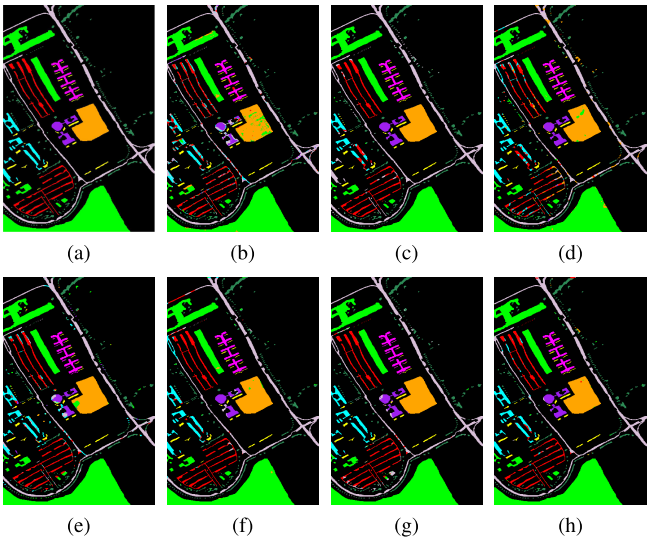


Fig. 6. Classification maps obtained via various algorithms for the UP dataset. (a) Ground truth. (b) 3-D-CNN. (c) SSRN. (d) HybridSN. (e) 3-D-GAN. (f) HyperGAMO. (g) HC-ADGAN. (h) CACS. Best zoomed-in view.

Table VI shows the classification effect on the KSC dataset. As can be seen from the accuracy highlighted, our CACS is superior to other compared algorithms in terms of MCs 2 and 3 and the suboptimal performance in MCs 4, 5, 6, and 10, so it also achieves the optimal results in the AA, OA, and K. It indicates that the combination of DGM based on Schmidt's mathematical principle and appropriate loss function can effectively extract the category-specific features in HSIs. In addition, SSRN obtained suboptimal OA, AA, and K values through an excellent feature extraction network and obtained the optimal performance on dominated categories 9, 11, 12, and 13, the best classification results in MC 10, and the suboptimal results in MCs 2 and 3. The stability performance on three different datasets demonstrates that the deep generative model involved in our network is effective in data augmentation of MCs and feature extraction with class specificity.

Figs. 6–8 depict the classification visual maps of different algorithms on the three datasets, and the ground truth is drawn again to facilitate comparing results. From the perspective of subjective observation, the classification effect of 3-D-CNN, HybridSN, and 3-D-GAN is poor, resulting in many visually noisy scattered points. The results of SSRN and HyperGAMO

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TABLE V
CLASSIFICATION ACCURACY OF DIFFERENT ALGORITHMS FOR THE LOUKIA DATASET

No.	3D-CNN [6]	SSRN [7]	HybridSN [8]	3D-GAN [15]	HyperGAMO [28]	HC-ADGAN [39]	CACS
01	42.34±7.80	60.84±10.99	51.35±12.74	51.17±12.89	**98.24±2.95**	56.94±6.92	51.82 ± 14.31
02	71.87±15.94	58.59±41.03	66.41±21.83	78.44±18.76	28.67±8.90	86.57±11.08	**97.27±3.56**
03	78.33±5.05	81.32±8.00	75.48±11.68	44.95±7.80	44.58±8.50	67.93±9.18	**86.24 ± 3.58**
04	10.00±6.43	12.53±12.97	28.67±10.40	18.29±6.15	**46.62±12.24**	41.49±17.75	42.00 ± 16.85
05	74.51±6.62	87.44±6.41	88.48±3.34	82.56±6.03	68.59±6.03	84.37±5.19	**89.87 ± 3.63**
06	29.57±7.99	31.65±7.57	30.57±9.06	20.66±4.47	**99.65±0.39**	22.87±5.38	47.64 ± 9.43
07	45.62±8.51	63.79±11.48	54.86±5.73	44.61±8.74	**93.68±2.97**	55.20±3.62	68.74 ± 4.57
08	69.86±4.75	79.64±14.46	68.20±15.14	78.83±7.75	64.42±12.37	**93.08±7.35**	83.30 ± 3.30
09	83.65±2.94	**87.87±4.49**	79.13±6.26	73.96±5.63	87.84±3.29	60.80±3.65	81.64 ± 3.29
10	81.27±2.16	79.69±4.60	75.90±5.89	79.47±2.86	72.47±3.38	78.52±2.61	**84.50 ± 3.08**
11	62.87±4.34	80.97±6.91	72.24±4.98	72.84±5.73	**89.00±3.24**	81.93±5.21	79.36 ± 3.35
12	91.88±2.91	80.15±19.64	90.11±5.43	89.37±3.58	**100.00±0.00**	89.73±4.38	90.15 ± 6.32
13	**100.00±0.00**	90.00±30.00	**100.00±0.00**	**100.00±0.00**	83.23±5.13	**100.00±0.00**	**100.00±0.00**
14	99.70±0.38	97.95±3.00	**99.93±0.21**	99.91±0.29	86.72±1.94	99.57±0.95	99.80 ± 0.32
AA (%)	67.25±2.34	70.89±3.88	70.10±3.40	66.79±2.73	75.98±2.41	69.47±2.35	**78.74 ± 1.19**
OA (%)	79.12±1.39	82.29±3.67	78.80±2.47	76.53±0.64	84.00±1.38	76.31±0.68	**84.31 ± 0.69**
$k \times 100$	74.92±1.68	78.78±4.51	74.71±2.91	72.08±0.78	80.83±1.69	72.52±1.35	**81.40 ± 0.79**

TABLE VI
CLASSIFICATION ACCURACY OF DIFFERENT ALGORITHMS FOR THE KSC DATASET

No.	3D-CNN [6]	SSRN [7]	HybridSN [8]	3D-GAN [15]	HyperGAMO [28]	HC-ADGAN [39]	CACS
01	99.45±0.44	98.92±3.25	96.84±5.31	96.62±2.94	97.67±5.23	97.63 ± 0.89	**99.68±0.55**
02	71.95±6.42	92.20±11.47	82.98±10.64	83.50±11.57	75.31±6.94	79.01±8.82	**92.21±6.37**
03	90.00±7.48	94.90±8.85	95.31±3.92	63.30±16.76	82.42±14.72	82.03±10.19	**95.72±4.00**
04	59.21±10.45	75.76±18.76	68.62±12.41	74.90±11.59	**90.23±9.16**	60.71±11.60	78.73±11.58
05	68.17±11.07	73.82±20.67	68.07±16.56	87.37±9.87	78.60±11.67	**99.38±13.47**	80.28±12.47
06	86.61±6.61	89.91±16.09	76.39±11.81	**92.63±7.85**	68.66±10.41	86.46±7.29	90.52±4.38
07	96.80±3.76	87.84±16.64	97.00±4.08	88.22±10.66	**99.02±1.73**	94.28±8.30	92.83±9.48
08	92.15±5.75	95.27±12.78	95.50±11.90	74.51±11.94	99.33±1.09	**99.62±0.98**	95.88±4.32
09	94.17±2.99	**100.00±0.00**	97.80±2.08	96.62±4.11	91.57±6.61	81.54±2.73	**98.75±2.79**
10	92.79±3.42	**99.97±0.08**	95.17±7.27	97.32±3.44	97.09±2.36	98.85±2.15	99.00±1.32
11	98.39±1.14	98.92±1.70	**99.81±0.54**	89.90±6.97	95.73±5.22	97.85±1.28	98.83±0.66
12	94.41±2.70	**99.51±1.13**	93.66±4.33	91.06±6.11	99.26±0.97	97.41±2.05	99.06±1.37
13	**100.00±0.00**	**100.00±0.00**	99.71 ± 0.44	99.93±0.15	99.90±0.19	**100.00±0.00**	**100.00±0.00**
AA (%)	88.01±1.51	92.85±3.15	89.76±2.98	87.35±4.18	90.37±1.62	84.01±1.82	**93.88±0.64**
OA (%)	92.05±1.11	96.03±2.16	93.27±2.33	90.41±3.16	93.73±1.34	92.88±1.66	**96.40±0.36**
$k \times 100$	91.14±1.24	95.57±2.40	92.51±2.59	89.30±3.52	93.01±1.50	92.07±1.53	**96.00±0.40**

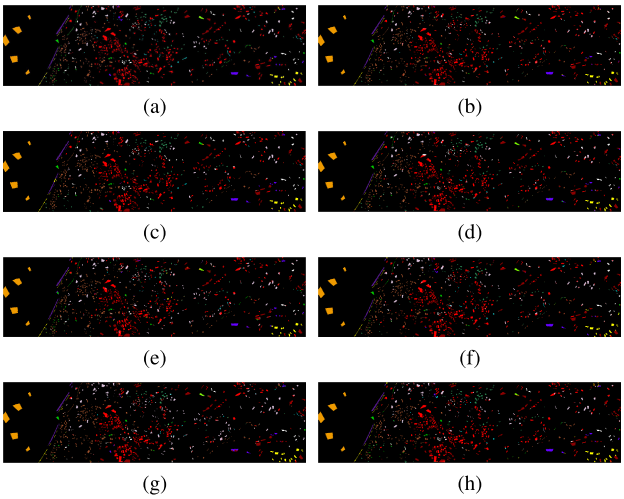


Fig. 7. Classification maps obtained via various algorithms for the Loukia dataset. (a) Ground truth. (b) 3-D-CNN. (c) SSRN. (d) HybridSN. (e) 3-D-GAN. (f) HyperGAMO. (g) HC-ADGAN. (h) CACS. Best zoomed-in view.

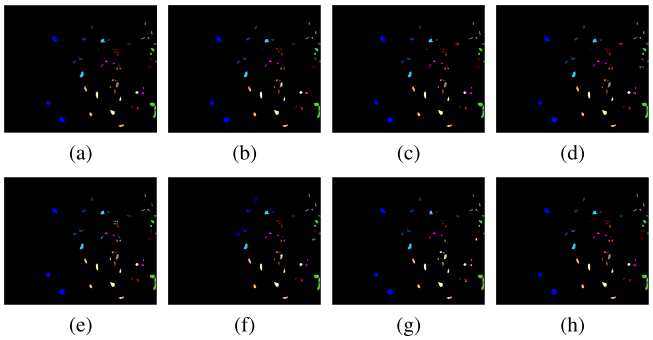
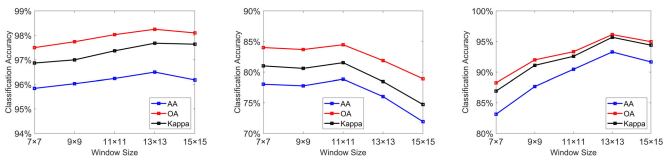


Fig. 8. Classification maps obtained via various algorithms for the KSC dataset. (a) Ground truth. (b) 3-D-CNN. (c) SSRN. (d) HybridSN. (e) 3-D-GAN. (f) HyperGAMO. (g) HC-ADGAN. (h) CACS. Best zoomed-in view.

classifications are better, and the error ratios of the majority classes are significantly reduced. Compared with other methods, CACS has an excellent classification effect on MCs. In addition, moreover, CACS performs better on confusing categories such as MC 3 (blue) and MC 8 (red) in the UP dataset.

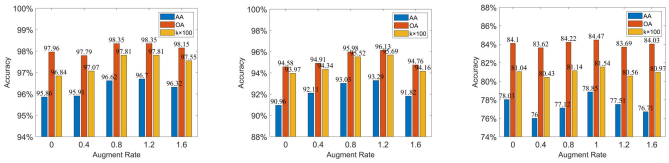
It can be inferred that this is because CACS expands the discriminability between different but similar categories

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(a) (b) (c)

Fig. 9. Classification accuracy with different window sizes. (a) UP. (b) Loukia. (C) KSC.



(a) (b) (c)

Fig. 10. Classification accuracy with different augmentation rate. (a) UP. (b) Loukia. (C) KSC.

in potential space, generates easy-to-classify features, and enhances the recognition ability for categories that are difficult to distinguish. In summary, these visual classification maps show that the proposed CACS can be better applied to practical applications with imbalanced HSIs.

D. Impact of Different Window Size

To highlight the advantages of our proposed method, we tune the hyperparameters of the network in this section. As mentioned in Section III-A, we first divide HSIs into 3-D cubes with sizes of W and input them into the 3-D encoder. Resizing the 3-D cubes can change the extent of the space in which the encoder extracts features. It is worth noting that the imbalanced classifier in the first stage is not weighted during the test, to fully explore the performance of window size feature extraction.

Fig. 9 describes the classification results AA, OA, and k for three datasets in the interval with W vary in $\{7 \times 7, 9 \times 9, 11 \times 11, 13 \times 13, 15 \times 15\}$. As can be seen from Fig. 9, for the UP and KSC datasets, the classifier performs best when W is set to 13×13. For the Loukia dataset, when W is 11×11, the experimental result is the best. It can be indicated that the results are better when W is moderate, because the spatial context information is difficult to fully utilize when it is small, and the large block of neighborhood information surrounding the central pixel may interfere with the classifier when it is too large.

E. Ablation Study

1) Augmentation Rate: The main innovation of our proposed CACS network is to generate features with category specificity based on Schmidt's mathematical theory. We change the proportion ar of generated samples to change the number of generated features so that the total number of features was ar $\times$ zdim, and the selection range of parameters was set as $\{0, 0.4, 0.8, 1.2, 1.6\}$. The test results on the UP dataset are shown in Fig. 10(a). It can be seen that 40% doubling and 120% doubling are the two conditions. The overall classification accuracy of the UP dataset is similar, but the accuracy of small categories such as 3, 4, 5, 7, and 9 is almost 1% higher, while the accuracy of large categories such as 1, 2, and 6 is slightly decreased. Therefore, for the UP dataset, ar is set to 1.2.

Fig. 10(b) shows the OA, AA, and k values of the Loukia dataset under different doubling ratios, and the classification accuracy is the highest at ar $=$ 1. For the KSC dataset, as shown in Fig. 10(c), the accuracy is the highest when ar $=$ 1.2. The classification results are better when ar $>=$ 1, especially the AA index, which can indicate the positive effect of the proposed deep generative model.

TABLE VII
RESULTS OF SCHMIDT-BASED AND WITHOUT SCHMIDT ON THREE DATASETS

Methods		UP	Loukia	KSC
Without Schmidt	AA	96.44±0.78	78.08±1.77	91.98±0.92
	OA	98.06±0.43	81.06±1.45	95.05±0.94
	k	97.43±0.57	77.63±1.64	94.49±1.05
Schmidt-based	AA	97.16±0.39	78.74±1.19	93.88±0.64
	OA	98.22±0.34	84.31±0.69	96.40±0.36
	k	97.64±0.44	81.40±0.79	96.00±0.40

TABLE VIII
RESULTS OF THE WEIGHTED AND UNWEIGHTED CLASSIFIER ON THE UP DATASET

No.	Unweighted classifier	Weighted classifier
01	98.34 ± 1.13	96.96 ± 1.78
02	99.91 ± 0.07	99.41 ± 0.53
03	89.89 ± 5.06	92.82 ± 3.61
04	96.42 ± 0.93	96.20 ± 1.50
05	98.87 ± 0.80	99.55 ± 0.36
06	98.78 ± 1.78	99.87 ± 0.18
07	89.54 ± 9.84	98.26 ± 1.73
08	97.22 ± 1.21	97.58 ± 1.99
09	93.61 ± 4.53	93.77 ± 3.82
AA (%)	95.84 ± 1.57	97.16 ± 0.39
OA (%)	98.07 ± 0.62	98.22 ± 0.34
$k \times 100$	97.43 ± 0.82	97.64 ± 0.44

2) Schmidt-Based Augment Features: To evaluate the contribution of the Schmidt-based augment features module, we replace the augment method with multiplying features of minority classes. We compare the Schmidt-based module with the module without Schmidt and the experimental results on the three datasets are shown in Table VII.

It can be seen from the results that feature augmenting based on Schmidt's theory can help improve the classification effect of MCs.

3) Weighting Classifier: In the first stage, we adopt the cost-sensitive loss function to weigh the imbalanced features of the input classifier according to the number of features and classification difficulty. In this section, we compare the influence of weighted and unweighted classifiers on results through experiments. Tables VIII–X, respectively, represent the classification accuracies of unweighted and weighted classifiers on the UP, Loukia, and KSC datasets.

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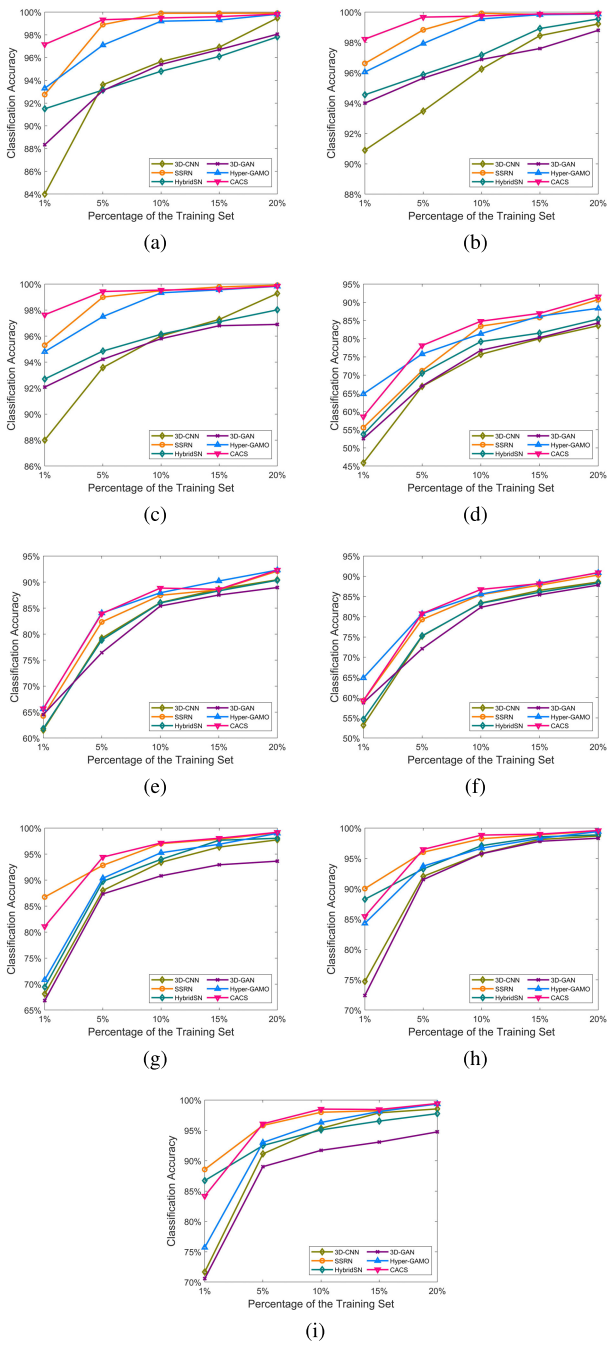


Fig. 11. Classification maps with different percentages of training samples. (a) UP-AA. (b) UP-OA. (c) UP-k. (d) Loukia-AA. (e) Loukia-OA. (f) Loukia-k. (g) KSC-AA. (h) KSC-OA. (i) KSC-k.

It can be seen that the AA values of the weighted results are significantly increased, which indicates that adaptive weighting of classifiers in the first stage can improve the classification result of MCs.

F. Influence of Different Percentages of Training Samples

To further investigate the classification performance of the CACS network, we will extract different proportions of samples from three datasets as training sets to conduct the above experiments. The UP, Loukia, and KSC datasets were extracted with 1%–20% of the total sample as the training set with an interval of 5%. As can be seen from Fig. 11, with the increase of the proportion of labeled data, the classification accuracy of CACS improves in all three datasets.

Compared with other algorithms, CACS consistently outperforms other comparison algorithms in any proportion of any dataset, which illustrates the generalization of the proposed method.

TABLE IX

RESULTS OF THE WEIGHTED AND UNWEIGHTED CLASSIFIERS ON THE LOUKIA DATASET

No.	Unweighted classifier	Weighted classifier
01	66.02 ± 8.07	51.82 ± 14.31
02	91.88 ± 9.69	97.27 ± 3.56
03	83.79 ± 3.48	86.24 ± 3.58
04	41.87 ± 14.38	42.00 ± 16.85
05	87.42 ± 3.95	89.87 ± 3.63
06	36.75 ± 5.55	47.64 ± 9.43
07	68.46 ± 5.62	68.74 ± 4.57
08	81.19 ± 5.22	83.30 ± 3.30
09	82.55 ± 2.20	81.64 ± 3.29
10	84.61 ± 2.29	84.50 ± 3.08
11	78.72 ± 3.79	79.36 ± 3.35
12	92.31 ± 1.87	90.15 ± 6.32
13	100.00 ± 0.00	100.00 ± 0.00
14	99.27 ± 0.90	99.80 ± 0.32
AA (%)	78.22 ± 1.19	78.74 ± 1.19
OA (%)	84.21 ± 0.46	84.31 ± 0.69
$k \times 100$	81.22 ± 0.53	81.40 ± 0.79

TABLE X

RESULTS OF THE WEIGHTED AND UNWEIGHTED CLASSIFIERS ON THE KSC DATASET

No.	Unweighted classifier	Weighted classifier
01	99.72 ± 0.44	99.63 ± 0.54
02	89.72 ± 9.57	90.86 ± 10.77
03	95.27 ± 3.52	96.48 ± 3.80
04	81.54 ± 7.83	82.80 ± 10.76
05	78.84 ± 13.93	79.81 ± 16.74
06	83.43 ± 11.03	90.52 ± 5.61
07	92.50 ± 7.76	91.00 ± 8.56
08	98.87 ± 0.97	95.98 ± 3.74
09	99.97 ± 0.07	100.00 ± 0.00
10	97.17 ± 3.24	99.31 ± 1.79
11	97.58 ± 2.84	99.44 ± 0.58
12	98.12 ± 2.47	97.56 ± 2.57
13	100.00 ± 0.00	100.00 ± 0.00
AA (%)	93.29 ± 1.85	94.11 ± 1.78
OA (%)	96.13 ± 1.01	96.63 ± 0.88
$k \times 100$	95.69 ± 1.13	96.24 ± 0.98

G. Influence of Alpha and Starting Epochs

In addition to the above parameters, the classifier in the first stage sets appropriate weights for different samples according to the influence degree of samples on the decision boundary and the number of samples belonging to the same category, to better classify imbalanced data. Specifically, to prevent CACS from overfitting, the classifier reduces the weight of samples that have a large impact on the decision boundary and increases the weight of MCs, where the parameters are

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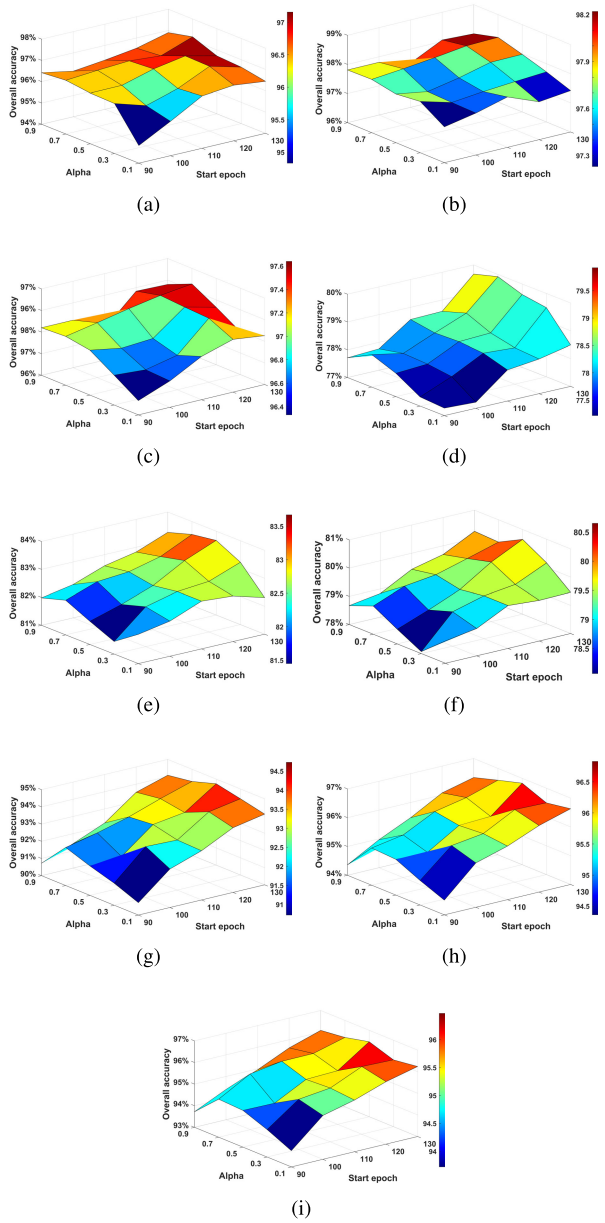


Fig. 12. Classification maps with different alphas and starting epochs. (a) UP-AA. (b) UP-OA. (c) UP-k. (d) Loukia-AA. (e) Loukia-OA. (f) Loukia-k. (g) KSC-AA. (h) KSC-OA. (i) KSC-k.

alpha and e. Among them, the hyperparameter alpha is used to adjust the degree of weight balancing, and the parameter e is the iteration from which the weight balance starts. That is, the classifier that classifies imbalanced data does not carry out category weighting at the beginning, but balances the weight of the classifier according to the overfitting degree of the network after iteration to the eth epoch. The number of alpha and e are varied in the grid of $\{0.1, 0.3, 0.5, 0.7, 0.9\}$ and $\{90, 100, 110, 120, 130\}$, respectively, as can be seen by the experiments in Fig. 12, the UP dataset gets the optimal result when alpha $=$ 0.7 and e $=$ 130. Although the weight balance of unbalanced classifiers does not significantly improve the results from the perspective of the OA, AA, and k, the classification accuracy of MCs is increased, while the classification accuracy of dominated categories is decreased.

TABLE XI
TRAINING AND TEST TIMES (IN SECOND) OF DIFFERENT METHODS ON THE THREE DATASETS

Methods		UP	Loukia	KSC
3D-CNN	Training time	131.94	251.73	117.18
	Test time	15.29	5.67	2.51
	Parameters	0.4464	0.7816	0.7814
	FLOPs	0.2375	0.4132	0.4130
SSRN	Training time	205.01	507.67	128.25
	Test time	24.88	8.82	4.51
	Parameters	1.5170	2.3617	2.3615
	FLOPs	4.1711	6.4951	6.4948
HybridSN	Training time	28.91	45.33	19.64
	Test time	11.31	3.10	1.36
	Parameters	9.1084	9.1158	9.1143
	FLOPs	11.9375	11.9472	11.9452
3D-GAN	Training time	194.59	399.06	146.01
	Test time	2.74	5.04	1.86
	Parameters	28.1754	28.1675	28.1673
	FLOPs	1066.1462	1066.1474	1066.1472
HyperGAMO	Training time	3662.38	7048.61	5499.09
	Test time	19.78	9.33	4.06
	Parameters	353.7305	585.4326	541.1111
	FLOPs	582.2420	969.4513	893.8243
CACS	Training time	263.13	534.57	136.40
	Test time	85.01	64.34	34.37
	Parameters	2.1724	3.3537	3.6925
	FLOPs	327.9961	352.2956	117.1492

This is because the classifier is weighted by category, which increases the attention to MCs while attenuating the attention to the majority of samples.

H. Computation Complexity of Different Algorithms

Finally, the computational complexity of different algorithms is analyzed in this section. By executing the comparison algorithms and CACS network on the same GeForce GTX 2080 GPU to classify the three general HSI datasets, the computational training time and testing time of all algorithms are counted, as shown in Table XI. It can be seen that the 3-D-CNN network has the lowest complexity. The training and testing times of the HybridSN are the shortest in the three datasets.

Because the HyperGAMO algorithm requires cyclic training in five submodules, the training time is the longest, and the number of parameters is the largest. Since the patch size inputting to the encoder is several times that of other algorithms and expands during the convolution process, the FLOPs of 3-D-CNN are much higher than those of other algorithms. Our CACS increases the training time appropriately in exchange for the highest classification performance. At the same time, to test accurately, the CACS uniformly multiplies the test data by ten times, so the test time is extra long.

V. CONCLUSION

In this article, we propose a CSCA model for imbalanced HSIC which consists of two stages. The first stage consists of a 3-D encoder, a 3-D decoder, and a custom imbalanced classifier. The second stage consists of a minority sample-generated module based on Schmidt's mathematical theory and a balanced classifier. By setting class-specific feature prototypes, CACS can extract features with large interclass differences

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and small intraclass differences. At the same time, the generation of a few class samples is not by simply copying the original features, but by Schmidt's mathematical principle to generate features containing class information and discriminability. In addition, we weighted the loss function of the first-stage classifier to optimize the classification performance and improve the accuracy and expressiveness of features. After ten independent repeated experiments with different seeds, the average classification effect of CACS reached the best. In the future, we will represent feature prototypes of each class through multiple vectors to improve the expression ability of the prototypes and optimize the distribution of generated samples.

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